Centralized Rineya Titan 0.6B: qualification cycle

Date: 28 September 2026. **Native work is blocked at host-RAM admission. No model was trained, no acquisition or retention effect was measured, and no recipe was promoted.**

Scope and exact identity

The primary student is **Rineya-Titan-Experimental-1.0.0-devfreeze**, measured at 596,049,920 parameters in 310 BF16 tensors. It has 28 layers, hidden size 1024, MLP 3072, 16 query heads, 8 KV heads, head dimension 128 and vocabulary 151936. Tied embeddings are declared; loaded pointer tying is still unmeasured. 40960 is a config declaration, not a qualified context envelope.

Weight SHA256: `537201676814ade103c4efa63f5ebe758ea309971803c4ad633dbca5b6b531b7`. Config SHA256: `660db3b73d788119c04535e48cf9be5f55bc3100841a718637ae695b442f27dd`. Tokenizer SHA256: `aeb13307a71acd8fe81861d94ad54ab689df773318809eed3cbe794b4492dae4`. Template SHA256: `a55ee1b1660128b7098723e0abcd92caa0788061051c62d51cbe87d9cf1974d8`.

Exact local checkpoint: `C:/Users/clclc/BusinessProject/Chat-Bubble/Rineya_Published_Cores/Rineya-Titan-Experimental-1.0.0-devfreeze`. All root checkpoint assets and the complete tensor inventory are recorded in `receipts/PRIMARY_STUDENT_RECEIPT.json` and `receipts/PARAMETER_INVENTORY.json`.

This choice follows the user's Titan preference and keeps a simple dense BF16 base. It is not a measured superiority claim over Titan-Sparse or Titan-Transition. Their FP32 composition/rounding factors stay outside the first causal chain. The 344M draft is archived and no 344M GPU work occurred.

Source and protocol

Freshly fetched parent: `03db218ceebc61a2bc04f2482cf7e094045e01c1`. Execution branch: `research/post-training-mimo-centralized-v1`, initially pushed at that parent. Registration commit: `577ec6d5000bad11eaa72823d354b143b4178a26`. Corrected E0 retry source: `285529d9ad328de49a3e820ebc8fbdb0d18d693e`. No parent/main merge or historical branch rewrite occurred. The checkout had untracked drafts at arrival; AMENDMENT_001 explicitly records that it was not fully clean and preserves their original bytes. Both native attempts launched with clean source.

The supplied 66-page report was read in full; its staged experimental logic was retained and its 344M resource assumptions were superseded. Source PDF SHA256: `a1c5b5c66e1cef0edd04631ab7c487c93731b1a207bb67ee87fddd40cf43c2d7`.

Measured qualification results

Run	Result	Model loads / updates	Measured cost
Static binding	Exact Titan files/header verified	0 / 0	Streamed hashes; wall not continuously metered
Final CPU controls	80 tests passed	0 / 0 native	25.732s reported test execution, excluding imports
E0 attempt1	WindowsPath API failure before model load	0 / 0	185.313s worker;191.422s supervisor;705,085,440B peak process-tree RSS
E0 attempt2	HOST_RAM_GUARD	0 / 0	67.797s worker; 73.891s supervisor; 722,173,952B peak process-tree RSS
Bridge preparation	4608 open parents verified	0 / 0	17.009s;0 teacher/model calls

Attempt 1 failed because Windows psutil.disk_usage rejected a pathlib.WindowsPath. The retry changed only conversion to string and added import progress events. No resource guard or numerical tolerance was relaxed.

Attempt 2 measured **6.911331GiB host RAM available, below the 7.4GiB admission floor**. The worker floor remains 6.0GiB. GPU free memory was 6.890625GiB against a 4.0GiB admission floor. Worker RSS at this point was 0.668579GiB. Runtime import took 66.391s. The detected GPU is RTX4070 Laptop with 8188MiB. These are pre-load measurements.

Runtime: Python 3.11.15, torch 2.10.0+cu126, Transformers 4.57.1, safetensors 0.7.0, tokenizers 0.22.2. No pretrained tensor allocation or optimizer update occurred. Every attempted run has config, source, runtime, failure, resource, cost and process containment receipts. The watchdog reaped its owned worker tree.

Scientific interpretation

This rules out launching the registered run at the observed host-RAM state. It does **not** establish that Titan cannot fit after admission, that BF16 cannot work, that LoRA cannot learn, or that any capability regresses. Frozen-load/forward/backward VRAM, optimizer/activation memory, throughput, checkpoint timing and resume remain UNMEASURED on this model. There is no native zero-adapter, likelihood or gradient pass.

Historical Titan behavior used FP32 DenseReference. Its attention arithmetic differs from stock BF16 Qwen3; those historical scores are RECORDED FROM PRIOR WORK and do not transfer to this proposed runtime. The unchanged base remains CURRENT_BEST_RECIPE only as the campaign starting point, with no claimed improvement.

Data and held-out boundaries

Original bridge populations: 4096 TRAIN parents, 256 TRAIN_SELECTION and256 DEV; each role balances arithmetic, symbolic registers, finite sets and constrained text. 512 CHECK parents have hash-only reservations; no CHECK prompts/targets were exported. FINAL remains unopened, with its confirmation sample size unresolved.

Canonical allocation precedes rendering/solving. All 4608 open positives pass an independent verifier and all 4608 wrong-answer mutations fail. There are zero duplicate selected canonical parents and zero cross-parent prompt collisions. Four duplicate candidate specifications were skipped before selection. Full source/file hashes, generator runtime and provenance fields are retained; raw generated rows are local and regenerate from the committed code/registration.

Limits: canonicalization covers declared commutation/order equivalences, not general semantic equivalence or pretraining/historical decontamination. TRAIN finite-set answers include 105/1024 empty sets and text includes 12/1024 empty strings. Arithmetic has 428 distinct TRAIN answers across 1024 parents. Report trivial-answer baselines and per-family/structure effects; shared answers alone do not prove contamination. These are deterministic bridge tasks, not MiMo trajectories.

Tokenizer measurement

The pinned stock Qwen2TokenizerFast run completed in 111.268s with no PyTorch import, model load or model call. All 4608 open-role examples pass source hashes, exact template/prefix/completion checks and positional masking. Maximum total length is 157 tokens; maximum completion including EOS is 38. No truncation or oversize rows occurred. TRAIN contains 414,554 nonpadding tokens and 49,272 supervised tokens per pass; all open roles total 466,567 tokens. Replay is not yet enrolled, so these are acquisition-bank counts, not the complete E1 dose. Receipt and derived-file hashes are in receipts/BRIDGE_TOKENIZATION_RECEIPT.json; raw token arrays stay local. CHECK/FINAL were not opened. Tokenizer fit does not establish model resource fit.

Analysis controls and auxiliary failure

DEV analysis uses paired parents, averages seeds within parents and bootstraps whole parents within family strata. Duplicate seeds cannot manufacture independent evidence. Family regressions and source-correct losses remain visible; zero source denominators return null. The registered defaults are 10000 resamples, seed 20260927. The helper is exploratory, DEV-only, and confers no automatic promotion.

A no-model control chooses each family's most common TRAIN target (lexicographic tie break) and scores DEV: arithmetic 2/64, text 1/64, sets 3/64, registers 0/64. These are trivial data controls, not Titan results; no CHECK/FINAL contents were opened.

Tokenizer attempt1 selected the shell's Hermes environment with Transformers 5.14.1 instead of the pinned 4.57.1 environment. Its no-PyTorch-import guard failed before row tokenization, with zero model loads and no derived output. Exact failure wall time was not captured; the preserved receipt bounds it at 150-457s. Initial progress wording suggesting all-row processing was underway was unsupported and is corrected in the receipt. The retry pins the interpreter and checks package versions before import; the no-PyTorch guard remains enabled.

Preservation gate

No existing bank is yet qualified as 1024 broad-preservation replay rows for this Titan. The proposed plan is 768 deterministic replay rows across retrieval, instruction-following and answerability, plus 256 reviewed language replay rows. Answerability twins share parents; rows are not independent parent counts.

The v3 train-only 14000-row preservation slice is a language replay candidate; it cannot serve as fresh DEV. The MaxRL replay_train 1600 rows are a reasoning-only alternative. Neither supplies a broad-preservation certificate. Fresh DEV, actual source-correct enrollment, language scoring and a genuinely longer-context sentinel remain necessary. Historical spent/selection/sealed banks retain their original access boundaries. See `evidence/RETENTION_DESIGN.md` for exact metadata/exclusions.

Research stack and next expenditure

68 mechanisms/support components were source-audited. Current Titan native requalification remains false throughout. Separate GRPO, MaxRL-estimator and NGU rows expose their different variance, clipping and KL conventions. GFlowRL additionally needs canonical reward-maximizing R0; ordinary SFT is not that control. Architecture and inference experiments remain independent tracks.

The next highest-information native experiment is the **same bounded E0**, once adequate host headroom is measured. Preserve 7.4GiB admission and 6.0GiB throughout; do not infer required additional headroom from the pre-load deficit alone. No automatic identical retry, quantization, 344M fallback or paid cloud launch is justified. While blocked, use the completed tokenizer/analysis controls to resolve historical exclusions and preservation design. E1 seed 101 remains blocked until E0, data/replay and evaluation gates pass. The larger multi-method research program is ongoing; this report closes only the present source/data/preflight cycle.

Continuation 002: replay preparation advances; native gate still blocked

A fresh end-of-turn OS check found 6.999 GiB free host RAM before imports, with no Python worker running. It remains below the 7.4 GiB post-import native admission floor. No new native attempt was launched. The unchanged Titan remains the primary and current best; there are still zero pretrained loads, updates or efficacy results.

AMENDMENT_002 froze fixed original retrieval, priority-instruction and answerability banks. Construction verified 768 TRAIN rows (640 parents) and 256 DEV rows (192 parents), including grouped answerability twins. All 1024 reference answers and wrong-answer controls passed. Construction took 14.495s; tokenizer-only profiling 69.459s. TRAIN contains 243,336 nonpadding and 11,814 supervised tokens. Maximum open-role length is 401 tokens; no truncation. CHECK contains 384 additional parent hash reservations only, and FINAL remains closed.

Exact TRAIN-source inspection found that 4,000 original v1 rows were incorrectly labeled v2 in the 14,000-row preservation slice. Origin was recovered through exact original file/record membership, preserving source bytes. Source-quality exclusions left 4,644 v2 candidates in 298 conservative template groups; these are not proved independent statistical parents. Frozen quotas selected 256 language candidates. Assistant reviewers read all selected conversations: 250 references were acceptable and six had unsupported referents or claims. AMENDMENT_003 resolved five with reviewed pre-ranked alternatives and one with the next fixed-order same-class cluster. No target was rewritten; originals and rejected candidates remain available. This is assistant source review, not human approval or native grader calibration.

The curated 256 language rows have 22,642 nonpadding and 2,460 supervised tokens, with a maximum of 122 total tokens. The full candidate sequence epoch now has **5,120 rows, 680,532 nonpadding tokens and 63,546 supervised tokens**. Its maximum TRAIN length is 398 tokens. These are measured candidate-data quantities, not training throughput, a launch grant, or evidence of preserved capabilities.

A scoped audit compared 5,632 open acquisition/deterministic-retention rows against 32,800 historical whole-chat fingerprints and current normalized prompts, finding no collisions. It did not test historical user-only equality, all Titan ancestry, cross-schema semantic equivalence or pretraining exclusion. No protected payloads were opened. These limits remain explicit rather than becoming a blanket clean-data claim.

Validation passed 129 aggregate CPU tests plus five subsequent resolver tests. A shared-process tokenizer test issue was fixed solely in test isolation, preserving production no-PyTorch guards. Designs for fresh language DEV/calibration and a separate 2048/4096 long-context sentinel are registered as PROPOSED_UNQUALIFIED. Neither short retrieval nor source-reference review substitutes for those native evaluation gates.

The runtime has no removable sklearn/scipy/pandas/audio/vision bundle: only four testing packages exceed its required dependency closure. Cloning it is not a substantiated RAM fix. The original preload RSS also includes CUDA setup, so it must not be described as pure import overhead. Native work needs a real resource-state change; the user was asked to free host memory while independent preparation continued.